



OV Shashank
Electrical Engineering
Indian Institute of Technology Bombay
Specialization: Microelectronics and VLSI

14D070021
Dual Degree (B.Tech+M.Tech.)
Male
DOB: 24/06/1997

Examination	University	Institute	Year	CPI / %
Graduation	IIT Bombay	IIT Bombay	2019	9.81

SCHOLASTIC ACHIEVEMENTS

- Currently **ranked 1st** based on CPI among 60+ students in the Electrical Engineering Department, IIT Bombay
- Completed a **Minor** in the **Biosciences and Bioengineering** Department with a Minor CPI of **10**
- Three-time recipient of **Institute Academic Prize** awarded to top two students of each department annually

POSITION OF RESPONSIBILITY

Graduate Teaching Assistant | *CMOS Analog VLSI Design* (July '18 - Present)

- Assisted the professor in managing **logistics, course plan** and ensuring smooth functioning of the course
- Responsible for conducting **tutorials, designing assignments and projects** and evaluating submissions of **~140 students**

Junior Design Engineer: Data Acquisition and CAN | *IIT Bombay Racing* (Aug '15 - July '16)

- Implemented **Controller Area Network (CAN)** protocol to improve reliability and reduce wiring harness in a team of 6
- Designed onboard data logging system and real-time **wireless remote** data analysis with a **dashboard GUI** system

RESEARCH EXPERIENCE

Accelerating HB Analysis for RF Circuits (July '18 - Present)
Guide: Prof. Sachin Patkar, Electrical Engineering, IIT Bombay *MTech Thesis*

- Achieved **~900 GFlops, 2x** what is obtained by the Intel MKL for Dense LU Decomposition on the **Intel Xeon Phi KNL**.
- Implementation of the **Harmonic Balance** Steady-State Circuit Simulation algorithm for high frequency **RF circuits**
- **Modeling** of the MOSFET RF Circuits and other non-linear elements using charge oriented **Modified Nodal Analysis**
- Optimization **Block LU Decomposition** kernel for Block Sparse Matrices using **structure** and **sparsity aware** tuning

HPC Exposure and Exploration: IJEP Internship Program (May '17 - July '17)
Guide: Prof. Reiji Suda, Department of Computer Science, University of Tokyo

- Among the **top 3** students selected from the IITs for the **fully-funded** Summer Internship at **UTokyo** in Computer Science
- Achieved a maximum of **~3.5 TFlops** of performance in **Dense Matrix Multiplication** on Nvidia **GTX 1080 GPU**
- Accelerated **beyond baseline, SpMV** for multiple sparse matrix storage formats on a **Many-Core NUMA** Intel Xeon

COURSE PROJECTS

VCO Design and Layout | *Radio Frequency Microelectronics Chip Design* (Mar '18 - May '18)

- Awarded the **1st Prize** in the **Layout** Design Competition judged by **industry experts** in a class of **~40 students**
- Implemented an LC based **current-reuse Voltage-Controlled Oscillator** architecture with a tuning range of 5 to 6 GHz
- Used a Capacitor Bank and **varactor** for frequency tuning and achieved a **low Phase Noise** of -100 dBc at 1MHz Offset

Integer Factorization using QFS for Cryptanalysis | *VLSI Design Lab* (Jan '18 - May '18)

- Studied and implemented the **Quadratic Field Sieve** Algorithm in C++ and an HLS tool: **Bluespec System Verilog**
- Analysed **Sparse Boolean** Matrix structure and coded an optimized **Matrix-Vector** multiplication (BMVM) in Bluespec
- Created a **High-Precision** Integer class in C++ to operate on large numbers **without dynamic memory** allocation

PODEM and Deductive Fault Simulation | *Testing and Verification of VLSI Circuits* (Feb '18 - Apr '18)

- Implemented **PODEM** combinational stuck-at fault test-vector generation algorithm in C++ with **Verilog netlist parsing**
- Used **Deductive Fault Simulation** to minimise the set of test vectors for maximum fault-coverage by using fault dropping

TECHNICAL SKILLS & INTERESTS

HDLs & Programming Interests C++, C, Python, MATLAB, VHDL, Bluespec System Verilog, CUDA, MPI, Assembly
Circuit Simulation, RF and Mixed Signal Design, Processor Design, and Digital Design

EXTRA CURRICULAR ACTIVITIES

- Volunteered in a **L^AT_EX** workshop jointly conducted by the PG Academic Council and IRSCP Dec '17
- Learning the **Japanese** Language: Completed 100 hours with A+ grade Aug '17 - Present
- Volunteered in the **NSS Summer Volunteering** Program by teaching at Computer Literacy Program May - Jun '15